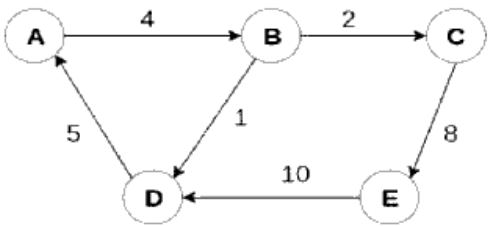


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Computer Engineering Semester :III Subject Code & Name: BTCOC303 Data Structures Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Write proper Syntax, example and program wherever necessary. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is a data structure? Why do we need data structures? Differentiate linear and non-linear data structure.	Remember	6
B)	Explain the concept of sparse matrices.	Understanding	6
C)	Explain double hashing in data structure with its advantages and disadvantages.	Understanding	6
Q.2	Solve Any Two of the following.		12
A)	What is Queue ADT? Explain representation and implementation of queue using sequential operations.	Synthesis	6
B)	Explain applications of stack for Expression Evaluation.	Understanding	6
C)	What is priority queue ? Explain operations of priority queue.	Analysis	6
Q. 3	Solve Any Two of the following.		12
A)	Explain circular linked list data structure with its insertion and deletion operations.	Analysis	6
B)	Write a C Program to implement following any two operations of doubly linked list. 1.insertion 2.deletion 3.display 4.search	Apply	6
C)	Justify a linked list is a data structure that is based on dynamic memory allocation. and List the application of Linked List Dynamic Memory Allocation.	Understanding	6
Q.4	Solve Any Two of the following.		12
A)	What is Binary Search Tree? Write an algorithm to search an element in Binary search tree.	Remember	6

B)	Explain Adjacency matrix for an undirected graph and what will be the adjacency matrix for the below directed weighted graph?  <pre> graph TD A((A)) -- 4 --> B((B)) B -- 2 --> C((C)) C -- 8 --> E((E)) E -- 10 --> D((D)) D -- 5 --> A </pre> Fig. Directed weighted graph	Synthesis	6
C)	Explain Threaded Binary Tree and its types? State its advantages and disadvantages.	Understanding	6
Q. 5	Solve Any Two of the following.		12
A)	What is a skip list? Write algorithm for basic skip list operations.	Remember	6
B)	Explain binary search algorithm by suitable example. Discuss the complexity of Binary search algorithm.	Analysis	6
C)	Explain Insertion sort algorithm with suitable example. Discuss the complexity of insertion sort.	Understanding	6
	*** End ***		

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